

Case Study**IT3060– Human Computer Interaction****Semester 2 2026****FitFlow Redesign Scenario:**

A mid-sized health-tech startup's fitness tracking app called FitFlow had been steadily losing users over the past year. Retention rates dropped significantly, and user ratings on major app stores declined from an average of 4.6 to 3.8 stars. The product team, concerned about the downward trend, decided to launch a comprehensive user research initiative. The goal was to uncover deep pain points, gather actionable insights, and validate new feature ideas including AI-powered personalized workout plans, social community features, and improved nutrition tracking. This effort evolved into a full-scale redesign project that followed a structured human-centered design process, addressing key learning outcomes from stakeholder identification through to final deployment.

The team started by thoroughly identifying stakeholders using a salience model that considered power, interest, and influence. Primary stakeholders included current and lapsed end users across diverse demographics, the internal product and design team led by Product Manager Priya Sharma, engineering and data science groups responsible for AI development, business leaders such as the CEO and marketing team focused on revenue recovery, and external partners like gym collaborations and regulatory bodies ensuring data privacy compliance. Secondary stakeholders encompassed family members who might engage with social features and healthcare providers interested in anonymized wellness trends. A dedicated stakeholder workshop helped align expectations and secure necessary resources and buy-in for the research phase.

Next, the team carefully selected appropriate research methods to ensure a balanced, mixed-methods approach. Quantitative methods included large-scale surveys distributed through Typeform and in-app prompts, detailed analysis of app analytics from Firebase and Mixpanel, retention cohort studies, and competitive benchmarking against rival fitness applications. Qualitative methods comprised semi-structured interviews with 35 users (both active and lapsed), focus group sessions, two-week diary studies, and contextual inquiries where researchers observed participants during actual workout sessions. To support these methods, the team designed several research instruments, including a detailed screening survey covering demographics, fitness habits, and satisfaction levels, a comprehensive interview guide with open-ended questions about daily usage and emotional experiences, a structured diary study template encouraging daily reflections with photo uploads, and clear consent forms emphasizing voluntary participation, data anonymization, and the right to withdraw at any time.

Ethical considerations guided every step of the user research. All participants received transparent information sheets and provided informed consent. Special attention was given to diversity in recruitment across age, gender, fitness levels, and geographic

regions, while maintaining strict data security protocols compliant with GDPR and CCPA standards. During the data collection phase, the team gathered rich qualitative insights through interview transcripts and diary entries, alongside quantitative data revealing that 68% of users abandoned the app shortly after onboarding and many found nutrition logging particularly tedious. This combination of qualitative stories and quantitative metrics provided a robust foundation for understanding user challenges.

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Following data collection, the team conducted thorough analysis of the research findings. They applied thematic analysis by first familiarizing themselves with all raw data, generating initial codes for recurring issues such as frustration with generic workout plans and desire for social accountability, then clustering these into major themes. Key themes that emerged included a significant personalization gap, feelings of social isolation within the app, high friction in daily tracking activities, and waning motivation due to missing celebratory elements. Affinity mapping workshops helped refine these themes, which were then presented to stakeholders in a detailed findings report complete with direct user quotes, journey visualizations, and supporting data charts.

Building on the analysis, the team created detailed user personas and empathy maps to humanize the insights. One prominent persona was Alex Rivera, a 32-year-old busy marketing professional who needed quick, adaptive workouts that fit irregular schedules but felt frustrated by rigid plans. Another was Priya Singh, a 27-year-old beginner teacher seeking safe social connections and accountability partners. Empathy maps for each persona captured what users said, thought, did, and felt, highlighting emotional pain points like demotivation and loneliness during solo fitness journeys. These artifacts helped the entire team maintain a user-centered perspective throughout subsequent design activities.

With clear user understanding established, the team moved on to writing user stories, mapping user journeys, and generating “How Might We” statements. User stories followed the agile format, such as “As a busy professional, I want AI-generated adaptive workout plans so that I can stay consistent despite my chaotic schedule.” Current-state user journey maps illustrated frustration peaks during onboarding and logging, while future-state maps envisioned smoother flows with intelligent recommendations and social integration. HMW statements guided ideation, including questions like “How might we make personalization feel truly intelligent and trustworthy?” and “How might we foster positive social connections without overwhelming users?”

These insights led directly to elucidating detailed requirements, both functional and non-functional. Functional requirements specified an AI workout engine, private social circles with robust privacy controls, camera-based nutrition logging powered by computer vision, and engaging progress dashboards. Non-functional requirements emphasized fast performance, offline capabilities, accessibility compliance, and strong data security. From these requirements, the team generated numerous interface ideas and developed low-fidelity wireframes in tools like Figma. Key screens included a home dashboard featuring a prominent AI “Daily Flow” recommendation card, a flexible workout builder with drag-and-drop elements, a dedicated social feed for challenges and group interactions, and a streamlined nutrition logger with a large camera button for instant food recognition. Wireframes included detailed annotations for user flows, microcopy, and interaction patterns.

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To validate these designs, the team planned and executed multiple rounds of usability testing. The test plan involved moderated remote sessions with representative users completing realistic tasks such as setting up an AI plan, logging meals, and joining a community challenge. Success metrics included task completion rates, System Usability Scale (SUS) scores, and qualitative feedback. Initial rounds revealed opportunities for improvement, such as adding more user control over AI suggestions and simplifying social navigation. Iterative testing across three rounds raised the SUS score from 68 to 87, confirming the designs effectively addressed core user needs.

With validated designs in hand, the team selected and justified an appropriate technology stack for implementation. React Native was chosen for the mobile frontend due to its cross-platform efficiency, rapid development capabilities, and strong support for complex animations needed in fitness experiences. The backend leveraged Node.js with Express and Firebase for real-time social features and scalable notifications. AI components utilized TensorFlow Lite for on-device personalization alongside cloud services for advanced models, with computer vision handled through accessible ML kits. This stack was justified by the startup's need for speed to market, cost-effectiveness, real-time capabilities, and privacy-focused architecture that minimized data transmission where possible.

Finally, the team prepared and deployed the redesigned mobile application. They built a Minimum Lovable Product incorporating the core new features, conducted extensive internal and beta testing with 150 users over several weeks, addressed bugs, optimized performance, and completed final accessibility and security audits. A phased soft launch in selected markets included A/B testing of key features, supported by a targeted marketing campaign emphasizing the app's intelligent AI coaching and rebuilt community aspects. Post-deployment monitoring allowed continuous improvements based on live user feedback and analytics. Within months, the redesign delivered measurable success: retention rates improved by 35%, average ratings increased to 4.5 stars, and premium subscription conversions rose significantly. This end-to-end project exemplified best practices across all targeted learning outcomes, transforming FitFlow from a declining product into a more engaging, user-loved fitness companion.